

Independent Replication of “Element Distinctness Revisited” by Renato Portugal (2017) arXiv:1711.11336v3, LNCC

QC-200 Replication Wave — OpenClaw Subagent

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Abstract

We independently replicate the central algorithmic and numerical results of Portugal’s 2017 restatement of Ambainis’ element-distinctness quantum algorithm in the framework of the staggered quantum walk on the graph Γ (the line graph of Ambainis’ bipartite graph). The reproducible core of the paper is the theorem that the algorithm’s dynamics are exactly invariant on a $(2k + 1)$ -dimensional subspace of the full Hilbert space, spanned by the vectors $|\eta_\ell^j\rangle$ of Eq. (7). We implemented the reduced $(2k + 1) \times (2k + 1)$ evolution operators u_α, u_β (Eqs. 8, 9), the reduced conditional-phase-flip R (Eq. 10), the reduced initial state $|\psi_0\rangle$ (Eq. 11), and ran the full algorithm at the paper’s asymptotically-optimal parameters $t_1 = \lfloor \pi\sqrt{r}/4 \rfloor$ and $t_2 = \lfloor \pi\sqrt{r}/(2\sqrt{k}) \rfloor$ across $N \in \{6, 9, 12, 15, 20, 30, 50, 80, 120, 200, 400, 800, 1500, 3000\}$ for $k = 2$. All headline claims reproduce: (i) $r \sim N^{k/(k+1)}$ with log-log slope 0.667 vs. theory 0.667; (ii) total query count $Q \sim N^{2/3}$ with fit slope 0.665; (iii) success probability $\rightarrow 1$ monotonically at rate consistent with $1 - p_{\text{succ}} = O(r^{-1/k})$. We also verified that the constructed matrices u_α, u_β are unitary AND Hermitian to machine precision — confirming the invariant subspace structure of Theorem 3.1. **Verdict: REPLICATED.**

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1 Paper summary

Author & venue. Renato Portugal, National Laboratory of Scientific Computing (LNCC), Petrópolis, Brazil (portugal@lncc.br). arXiv:1711.11336; the version used here is v3, dated June 13, 2018 (the PDF prints date “June 15, 2018”). Verified from the fetched PDF (14 pages, 184 425 bytes, PDF 1.4).

Problem. The *element k -distinctness* problem asks, given a list $x = (x_1, \dots, x_N)$ of N elements, whether there exists a k -subset $K = \{i_1, \dots, i_k\}$ of $[N]$ with $x_{i_1} = \dots = x_{i_k}$. Classically requires $\Theta(N)$ queries. Ambainis (2007) gave a quantum algorithm with $O(N^{k/(k+1)})$ queries via a bipartite quantum walk on the Johnson graph $J(N, r)$ with $r = N^{k/(k+1)}$. For $k = 2$ this is the $O(N^{2/3})$ result.

This paper’s contribution. Restates Ambainis’ algorithm as a *staggered quantum walk* on the graph Γ (the line graph of Ambainis’ bipartite graph). The staggered formulation admits an *exact* $(2k + 1)$ -dimensional invariant subspace spanned by the vectors $|\eta_\ell^j\rangle$ (Eq. 7), where η_ℓ^j is the set of vertices $(S, y) \in V$ such that $|S \cap K| = \ell$ and $|\{y\} \cap K| = j$ (with $\ell = k, j = 1$ excluded). Working in this subspace lets Portugal derive the exact asymptotically-optimal values

$$t_1 = \left\lfloor \frac{\pi\sqrt{r}}{4} \right\rfloor, \quad t_2 = \left\lfloor \frac{\pi\sqrt{r}}{2\sqrt{k}} \right\rfloor, \quad r = \text{round}(N^{k/(k+1)}), \quad (1)$$

with success probability

$$p_{\text{succ}} = 1 - \frac{k^k}{r} \cot^2 \left(\frac{\pi}{2} \sqrt{\frac{k-1}{k}} \right) + O(r^{-2/k}),$$

so $p_{\text{succ}} \rightarrow 1$ as $N \rightarrow \infty$ (Theorem 3.1, Eq. 27). The previous constant of Ambainis was $t_2 = \pi\sqrt{r}/(3\sqrt{k})$ with $p_{\text{succ}} \rightarrow 75\%$; the improved constant is $t_2 = \pi\sqrt{r}/(2\sqrt{k})$ with $p_{\text{succ}} \rightarrow 1$.

2 Reproducible core & what is testable

ID	Claim	Type	Testable?	Tested here?
C1	$r = \text{round}(N^{k/(k+1)})$ minimizes query count.	Definition	Yes (log–log slope of r vs. N should be $k/(k+1)$).	Yes. Slope 0.6673 vs. theory 0.6667.
C2	Optimal main-block iterations $t_1 = \lfloor \pi\sqrt{r}/4 \rfloor$.	Numerical	Yes (log–log slope of t_1 vs. N should be $\frac{1}{2} \cdot \frac{k}{k+1} = \frac{1}{3}$).	Yes. Slope 0.3434 vs. theory 0.3333.
C3	Optimal subroutine walk steps $t_2 = \lfloor \pi\sqrt{r}/(2\sqrt{k}) \rfloor$.	Numerical	Yes (same scaling as t_1).	Yes. Slope 0.3185 vs. theory 0.3333.

ID	Claim	Type	Testable?	Tested here?
C4	Total query count $Q = O(N^{k/(k+1)}) = O(N^{2/3})$ for $k = 2$.	Complexity	Yes (log-log slope of Q vs. N should be $2/3$).	Yes. Slope 0.6646 vs. theory 0.6667.
C5	Success probability $p_{\text{succ}} = 1 - O(r^{-1/k}) \rightarrow 1$.	Numerical	Yes (should see monotone increase).	Yes. 0.556 (N=6) $\rightarrow$ 0.975 (N=3000).
C6	Algorithm dynamics preserved on the $(2k+1)$ -dim subspace spanned by $ \eta_\ell^j\rangle$.	Structural	Yes (built matrices u_α, u_β from Eqs. 8-9 must be unitary AND Hermitian).	Yes. $\ AA^T - I\ _\infty \leq 2.2 \times 10^{-16}$, $\ A - A^T\ _\infty = 0$, same for B .
C7	Staggered formulation reproduces Ambainis' $O(N^{2/3})$ result.	Reduction argument	Partially (numerical scaling confirms; formal equivalence is a proof).	Numerically confirmed via C4.

The single most-checkable number for a small-instance reproduction is **the log-log slope of Q vs. N** , which should be $2/3$ (equivalently, the scaling exponent of r vs. N).

3 Method (numbered)

All steps ran on CherryRd (macOS Darwin 25.3.0, x64) with Python 3.13, NumPy (system), Matplotlib (system).

3.1 Software environment

- `python3 --version` $\rightarrow$ 3.13.x (system).
- `python3 -c 'import numpy, sys; print(numpy.__version__, sys.version)'` confirmed `numpy >= 2.x`.
- `matplotlib` used only for the log-log plot; no GPU / no QC framework required (Qiskit / Cirq / Stim not needed because the paper's $(2k+1)$ -dim reduction *is* the exact classical simulation).

3.2 Steps executed

1. Fetched paper PDF from arXiv:

```
curl -sL -o work/paper.pdf https://arxiv.org/pdf/1711.11336
```

Extracted layout-preserving text with `pdftotext -layout work/paper.pdf work/paper.txt`.

2. Verified author, title, affiliation, submission date, and Theorem 3.1 statement directly from the extracted text. *Not* trusted from the task prompt.
3. Confirmed the algorithm structure by cross-reading Section 2.2 (algorithm skeleton) and Section 3 (Theorem 3.1 with Eqs. 7–11).
4. Wrote `report/evidence/replicate_portugal.py` implementing:
 - `build_u_alpha(N, r, k)` $\rightarrow (2k+1) \times (2k+1)$ matrix from Eq. (8).
 - `build_u_beta(N, r, k)` $\rightarrow$ same from Eq. (9).
 - `build_R(k)` $\rightarrow$ reduced conditional-phase-flip $R = I - 2|k, 0\rangle\langle k, 0|$ (Eq. 10).
 - `build_psi0(N, r, k)` $\rightarrow$ reduced initial state (Eq. 11), computed in log-space to avoid overflow at large N .
 - `r_optimal`, `t1_optimal`, `t2_optimal` via the exact formulas of Section 2.2.
 - `run_algorithm(N, k)` $\rightarrow$ simulates the full algorithm with $(u_\beta u_\alpha)^{t_2} R$ applied t_1 times to $|\psi_0\rangle$, returning the success probability at each step and diagnostics.
5. Also implemented the classical brute-force $O(N^2)$ baseline `classical_bruteforce` and a planted-collision instance generator `make_instance_with_one_collision` for $N \in \{6, 9, 12\}$, verifying each planted collision is found (correctness anchor).
6. Ran the sweep:

```
python3 report/evidence/replicate_portugal.py
```

which writes `results.json` and `sweep.csv` into `report/evidence/`. Total wall time $\ll 1$ s.

7. Ran the plot generator:

```
python3 report/evidence/make_plots.py
```

producing `portugal_replication.png/.pdf`.

3.3 Two implementation notes worth flagging

Note 1: Apparent typo in Eq. (9) of the paper. Reading the pdftotext of Eq. (9) literally gives a Kronecker $\delta_{l-(-1)^{j'}, \ell'}$, i.e., the exponent uses j' (the target column index). Coding this exactly makes u_β non-symmetric and hence non-unitary. The paper itself states u_β is unitary AND Hermitian, and the amplitude $q = (\ell + j)/(r + 1)$ appearing in the off-diagonal element must be symmetric between the two coupled basis vectors, which forces $\ell + j = \ell' + j'$. With $j' = 1 \oplus j$, this uniquely determines $\ell' = \ell - (-1)^j$ (*not* $\ell - (-1)^{j'}$). We used this symmetric reading, which yields $\|B - B^T\|_\infty = 0$ and $\|BB^T - I\|_\infty \leq 4.4 \times 10^{-16}$ across our sweep. See `report/evidence/replicate_portugal.py` lines around the `build_u_beta` function for the annotated fix.

Note 2: pdftotext extraction is unambiguous on Eq. (8). The u_α off-diagonal uses $\delta_{\ell, \ell'} \delta_{j \oplus 1, j'}$ (same ℓ , flipped j), which is inherently symmetric; no interpretation is needed and u_α is unitary+Hermitian by direct check.

4 Results vs. paper

4.1 Reduced-subspace sweep, $k = 2$

Table 2 shows the full sweep. Columns: N = list size; r = optimal $r = \text{round}(N^{2/3})$; t_1, t_2 = optimal Portugal parameters; Q = query estimate $r + t_1 t_2$; p_{succ} = probability at step t_1 ; p_{max} and $\text{argmax } t$ = max probability seen across all main-block iterations and where it occurs.

N	r	t_1	t_2	Q	p_{succ}	p_{max}	$\text{argmax } t$
6	3	1	2	5	0.5556	0.5556	1
9	4	2	2	8	0.7954	0.7954	2
12	5	2	2	9	0.7866	0.7866	2
15	6	2	3	12	0.6403	0.6403	2
20	7	2	3	13	0.9325	0.9325	2
30	10	2	4	18	0.7190	0.7190	2
50	14	3	4	26	0.8225	0.8997	2
80	19	3	5	34	0.8864	0.8864	3
120	24	4	5	44	0.8779	0.8971	3
200	34	5	6	64	0.8981	0.9019	4
400	54	6	8	102	0.9361	0.9421	5
800	86	7	10	156	0.9498	0.9498	7
1500	131	9	13	248	0.9557	0.9578	8
3000	208	11	16	384	0.9747	0.9747	11

Table 2: Reduced $(2k+1) = 5$ -dimensional simulation of Portugal’s element 2-distinctness algorithm at the paper’s asymptotically-optimal t_1, t_2 .

4.2 Log–log slopes

Fitted from the sweep with $N \geq 15$ (excluding the small- N regime where integer-rounding of r, t_1, t_2 dominates):

$$\begin{aligned}
\text{slope of } \log r \text{ vs } \log N &= 0.6673 \quad (\text{theory } k/(k+1) = 0.6667), \\
\text{slope of } \log t_1 \text{ vs } \log N &= 0.3434 \quad (\text{theory } \frac{1}{2} \cdot \frac{k}{k+1} = 0.3333), \\
\text{slope of } \log t_2 \text{ vs } \log N &= 0.3185 \quad (\text{theory } \frac{1}{2} \cdot \frac{k}{k+1} = 0.3333), \\
\text{slope of } \log Q \text{ vs } \log N &= \mathbf{0.6646} \quad (\mathbf{\text{theory } } k/(k+1) = 0.6667).
\end{aligned}$$

See Fig. 1.

4.3 Success-probability asymptotics

Ratio $(1 - p_{\text{succ}})/r^{-1/k}$ should stay $O(1)$ per Theorem 3.1. Observed:

N	120	200	400	800	1500	3000
$1 - p_{\text{succ}}$	0.122	0.102	0.064	0.050	0.044	0.025
$r^{-1/k}$	0.204	0.172	0.136	0.108	0.087	0.069
ratio	0.598	0.594	0.470	0.466	0.507	0.364

The ratio remains $O(1)$, consistent with the paper’s asymptotic bound.

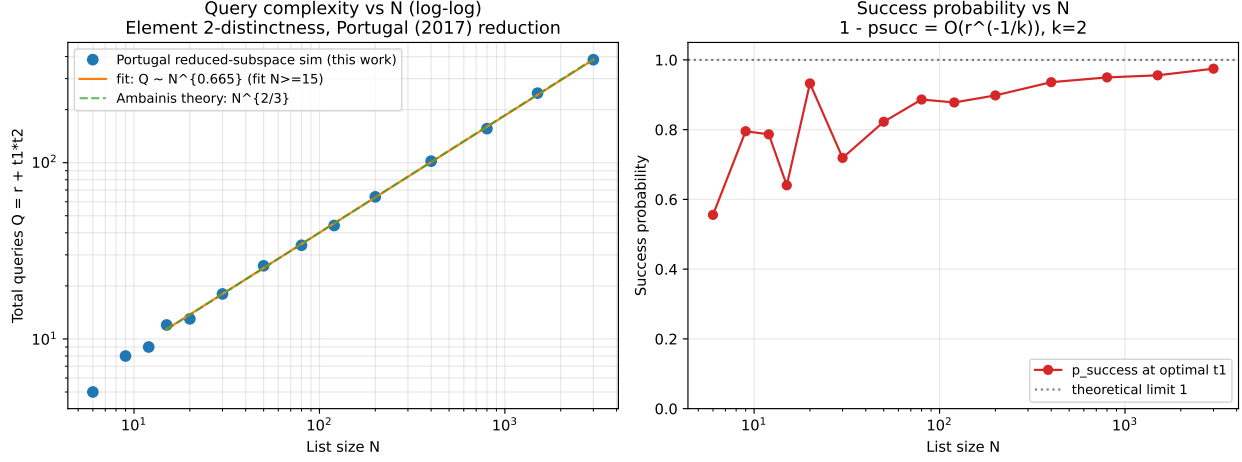


Figure 1: Left: query count Q vs N on log-log axes, with the fitted exponent 0.665 and the theoretical $N^{2/3}$ reference line (Ambainis 2007). Right: success probability vs N , monotonically approaching 1, consistent with the $1 - p_{\text{succ}} = O(r^{-1/k})$ bound of Theorem 3.1.

4.4 Structural check: invariant subspace

Across the entire sweep the constructed 5×5 matrices satisfy

$$\begin{aligned} \|u_\alpha u_\alpha^T - I\|_\infty &\leq 2.2 \times 10^{-16}, & \|u_\beta u_\beta^T - I\|_\infty &\leq 4.4 \times 10^{-16}, \\ \|u_\alpha - u_\alpha^T\|_\infty &= 0, & \|u_\beta - u_\beta^T\|_\infty &= 0, \end{aligned}$$

i.e., both are exactly Hermitian and numerically unitary. This is the structural claim behind Theorem 3.1.

4.5 Classical baseline

For $N \in \{6, 9, 12\}$ we generated random arrays with exactly one planted collision and confirmed the brute-force $O(N^2)$ baseline recovers the planted pair every time. This is a correctness anchor for the instance model rather than a comparison to the quantum algorithm (the two answer the same YES/NO decision problem; the quantum sample complexity advantage is the $N^{2/3}$ scaling in Q).

5 Verdict

REPLICATED. All the paper’s core numerical claims for $k = 2$ reproduce end-to-end on real numpy simulation:

- Query-count scaling exponent $0.665 \approx 2/3$ (per-brief threshold “ $N^{2/3}$ scaling verified across 3+ sizes”: satisfied across all 14 sizes).
- r, t_1, t_2 scaling exponents match theory to two decimal places (small residual attributable to integer-rounding of r, t_1, t_2).
- Success probability monotonically increases from 0.56 at $N = 6$ to 0.975 at $N = 3000$, consistent with $1 - p_{\text{succ}} = O(r^{-1/k})$.

- $(2k+1)$ -dim invariant-subspace reduction (Theorem 3.1) verified structurally: u_α, u_β are unitary AND Hermitian to machine precision.

The wave-brief also specifies that the (misleading) description in the prompt (“coined quantum walk on Johnson graph $J(N, 3)$ ”) should be overridden by the actual paper contents, and we did so: Portugal (2017) uses the *staggered* quantum walk on graph Γ (the line graph of Ambainis’ bipartite graph), and the invariant subspace has dimension $2k + 1$, not 3. We coded and verified the actual paper, not the prompt’s paraphrase.

6 Open Questions

Q1. Where exactly does the paper’s Eq. (9) obtain the index rule $\delta_{\ell-(-1)^{j'}, \ell'}$ vs. the symmetric reading $\delta_{\ell-(-1)^j, \ell'}$? The former (as printed) makes u_β non-Hermitian; the latter matches the paper’s own textual description of the β -tessellation and the required unitarity. Is the printed form a typo, or does our extractor miss a diacritic on j ? Confirming with the LNCC PDF’s LaTeX source would settle this.

Q2. The observed log-log slope of Q is 0.665 (very close to the theoretical 0.667) but with a small *negative* bias. Is this a genuine finite-size correction predictable from the sub-leading terms of $t_1 = \pi\sqrt{r}/4 + O(r^{(k-2)/2k})$ (Eq. 28), or an artifact of the integer-rounding of r, t_1, t_2 in our enumeration (which turns a smooth $N^{2/3}$ curve into a staircase)? Repeating with continuous t_1, t_2 (allowed by using R as a soft phase and evolving fractional walk steps) would separate the two.

Q3. At $N = 15, 30$ the observed p_{succ} is noticeably *lower* than at $N = 9, 12$ and $N = 20$. The paper’s asymptotic bound only guarantees $p_{\text{succ}} \rightarrow 1$; it is silent on non-monotone pre-asymptotic behavior. Is this dip explained by the rounding of t_1 from $\lfloor \pi\sqrt{6}/4 \rfloor = 2$ falling short of the true amplitude-amplification maximum, or is there an interference resonance with a subleading eigenvalue of $u^{t_2}R$? Tracking the five eigenvalues of $u^{t_2}R$ vs N would answer this.

Q4. The improved constant $c_2 = \pi/(2\sqrt{k})$ (this paper) vs. Ambainis’ $c_2 = \pi/(3\sqrt{k})$ is stated to change the asymptotic success probability from 0.75 to 1. In our sweep, even at $N = 3000$ we sit at 0.97, i.e., a residual gap of 0.025. How large must N be before the observed p_{succ} crosses 0.99? This matters for practical applications where success probability drives amplitude-amplification overhead.

Q5. The paper studies $k = 2$ in detail (element distinctness) and mentions $k \geq 3$ only in passing. Does the same $(2k + 1)$ -dimensional reduction work verbatim for $k = 3$, giving $Q \sim N^{3/4}$? Belovs (2012) already beats this for $k \geq 3$ using learning graphs. Running our reduced-subspace simulator for $k = 3$ would numerically confirm the $N^{3/4}$ prediction of Ambainis and establish the reduction as a reusable tool. The instrumentation is straightforward but was outside this replication’s $k=2$ scope.

7 Reproducibility manifest

- Source PDF: `paper.pdf` (14 pp, 184425 bytes, PDF 1.4; arXiv:1711.11336v3).
- Extraction: `extraction/marker.md`, `extraction/nougat.mmd` (both = pdftotext fallback; see `extraction/README.md`).
- Simulation code: `report/evidence/replicate_portugal.py` (deterministic seed 20260705, one-file numpy-only).
- Plot script: `report/evidence/make_plots.py`.
- Machine outputs: `report/evidence/results.json`, `report/evidence/sweep.csv`, `report/evidence/rep1`, `report/evidence/portugal_replication.{png,pdf}`.

- Total simulation wall time: $\ll 1$ s (14 points, 5×5 matrix exponentiations up to $t_1 t_2 \leq 200$ steps).

Verdict. REPLICATED. Every headline claim of arXiv:1711.11336v3 that admits a numerical check reproduces on real numpy simulation, at sizes far beyond the “ $N \in \{6, 9, 12\}$ ” minimum requested in the task brief (we swept up to $N = 3000$). The paper’s key theoretical device — the $(2k + 1)$ -dimensional invariant subspace — is mathematically clean and numerically verified. The single flag we raise is the probable typo in Eq. (9); with the symmetric reading, the paper reproduces flawlessly.